

Module 10) Rest Framework

1. Introduction to APIs Theory:

• What is an API (Application Programming Interface)?

Ans:

- An API is a set of rules that allows different software applications to communicate with each other.
- It acts as a bridge between the client (frontend) and the server (backend).
- APIs allow data exchange in formats like JSON or XML.

• Types of APIs: REST, SOAP.

Ans:

REST (Representational State Transfer):

- Lightweight and fast
- Uses HTTP methods like GET, POST, PUT, DELETE
- Data is mostly exchanged in JSON format

SOAP (Simple Object Access Protocol):

- Heavy and strict protocol
- Uses XML only
- More secure but slower than REST

• Why are APIs important in web development?

Ans:

- Enable communication between frontend and backend
- Allow integration with third-party services
- Support mobile apps and microservices
- Improve scalability and reusability of code

2. Requirements for Web Development Projects Theory:

- **Understanding project requirements.**

Ans:

- Identifying project goals and features
- Understanding user needs and data flow
- Deciding technology stack (Django, DRF, database)
- Planning API endpoints and security

- **Setting up the environment and installing necessary packages.**

Ans:

- Creating a virtual environment
- Installing Django and Django REST Framework
- Installing database drivers and required libraries
- Configuring settings and dependencies

3. Serialization in Django REST Framework Theory:

- **What is Serialization?**

Ans:

- Serialization is the process of converting Python objects into JSON format.
- Deserialization converts JSON back into Python objects.
- It allows data transfer over APIs.

- **Converting Django QuerySets to JSON.**

Ans:

- QuerySets cannot be sent directly over APIs
- DRF serializers convert QuerySets into JSON
- Makes data readable by frontend applications

- **Using serializers in Django REST Framework (DRF).**

Ans:

- Serializers define how model fields are converted to JSON
- They handle validation and data transformation
- Common types: ModelSerializer and Serializer

4. Requests and Responses in Django REST Framework Theory:

- **HTTP request methods (GET, POST, PUT, DELETE).**

Ans:

- GET: Retrieve data
- POST: Create new data
- PUT: Update existing data
- DELETE: Remove data

- **Sending and receiving responses in DRF.**

Ans:

- DRF uses Response objects
- Data is returned in JSON format
- Status codes indicate request success or failure

5. Views in Django REST Framework Theory:

- **Understanding views in DRF: Function-based views vs Class-based views.**

Ans:

Function-Based Views (FBV):

- Simple and easy to understand
- Less reusable

Class-Based Views (CBV):

- More structured and reusable
- Supports mixins and generic views

6. URL Routing in Django REST Framework Theory:

- **Defining URLs and linking them to views.**

Ans:

- URLs map API endpoints to views
- Defined in urls.py
- Routers simplify URL configuration in DRF

7. Pagination in Django REST Framework Theory:

- **Adding pagination to APIs to handle large data sets.**

Ans:

- Pagination divides large data into smaller chunks
- Improves performance
- Prevents server overload
- DRF provides PageNumberPagination and LimitOffsetPagination

8. Settings Configuration in Django Theory:

- **Configuring Django settings for database, static files, and API keys.**

Ans:

- Database configuration (MySQL, PostgreSQL, SQLite)
- Static and media file handling
- API keys and environment variables
- REST framework settings

9. Project Setup Theory:

- **Setting up a Django REST Framework project.**

Ans:

- Create Django project and app
- Install and configure DRF
- Define models and serializers

- Create views and URLs
- Test APIs using Postman

10. Social Authentication, Email, and OTP Sending API Theory:

- **Implementing social authentication (e.g., Google, Facebook) in Django.**

Ans:

- Allows login using Google or Facebook
- Uses OAuth 2.0 protocol
- Improves user experience
- Reduces password management

- **Sending emails and OTPs using third-party APIs like Twilio, SendGrid.**

Ans:

- Email APIs like SendGrid
- SMS APIs like Twilio
- OTPs enhance security
- Used for verification and password reset

11. RESTful API Design Theory:

- **REST principles: statelessness, resource-based URLs, and using HTTP methods for CRUD operations.**

Ans:

- Stateless communication
- Resource-based URLs
- Uses HTTP methods for CRUD
- Supports scalability and simplicity

12. CRUD API (Create, Read, Update, Delete) Theory:

- **What is CRUD, and why is it fundamental to backend development?**

Ans:

- CRUD stands for Create, Read, Update, Delete
- Forms the core of backend operations
- Enables full data management in applications

13. Authentication and Authorization API Theory:

- **Difference between authentication and authorization.**

Ans:

- Authentication: Verifies user identity
- Authorization: Controls user access

- **Implementing authentication using Django REST Framework's token-based system.**

Ans:

- Uses tokens instead of sessions
- Tokens are sent in request headers
- Secure and suitable for APIs

14. OpenWeatherMap API Integration Theory:

- **Introduction to OpenWeatherMap API and how to retrieve weather data.**

Ans:

- Provides real-time weather information
- Requires API key
- Returns data in JSON format
- Used for forecasting and climate apps

15. Google Maps Geocoding API Theory:

- **Using Google Maps Geocoding API to convert addresses into coordinates.**

Ans:

- Converts address into latitude and longitude
- Used for maps and location-based services
- Requires Google API key

16. GitHub API Integration Theory:

- **Introduction to GitHub API and how to interact with repositories, pull requests, and issues.**

Ans:

- Fetch repositories and commits
- Manage pull requests and issues
- Useful for DevOps and analytics tools

17. Twitter API Integration Theory:

- **Using Twitter API to fetch and post tweets, and retrieve user data.**

Ans:

- Allows posting tweets programmatically
- Fetches user and timeline data
- Used in social media analytics

18. REST Countries API Integration Theory:

- **Introduction to REST Countries API and how to retrieve country-specific data.**

Ans:

- Provides country details like population and currency
- No authentication required
- Used in geography and travel apps

19. Email Sending APIs (SendGrid, Mailchimp) Theory:

- **Using email sending APIs like SendGrid and Mailchimp to send transactional emails.**

Ans:

- Used to send transactional emails
- Reliable and scalable
- Supports email templates and analytics

20. SMS Sending APIs (Twilio) Theory:

- **Introduction to Twilio API for sending SMS and OTPs.**

Ans:

- Sends SMS globally
- Used for OTP verification
- Secure and fast communication

21. Payment Integration (PayPal, Stripe) Theory:

- **Introduction to integrating payment gateways like PayPal and Stripe.**

Ans:

- Enable online payments
- Support cards and digital wallets
- Secure transaction handling
- Widely used in e-commerce

22. Google Maps API Integration Theory:

- **Using Google Maps API to display maps and calculate distances between locations.**

Ans:

- Displays interactive maps
- Calculates routes and distances
- Used in logistics and navigation apps